Flexible Work Arrangements and Firm Outcomes: Experimental Evidence from India

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Abstract

This paper studies the impact of flexible work arrangements on worker and firm output in the context of a large apparel manufacturing cluster in India. Production is organized as batch processing lines, where tasks are performed sequentially to produce the finished product. In the status quo work arrangement, workers choose their own work timings, resulting in significant dispersion in arrival times at the firm. I implement an experiment with 96 small factories and 600 workers, where all workers in treatment firms are incentivized to arrive at work by the owners' preferred start time for a period of 4-weeks. Workers are paid a daily bonus equivalent to 20% of average earnings if they arrive on or before the start time. Incentives increase the likelihood that workers arrive on or before the preferred start time in treatment firms by 43% relative to control firms. This results in large improvements in coordination at the firm level – dispersion in worker arrival times decreases by 29%. There is no effect on worker attendance and a small increase in daily work hours of 16 minutes. Individual worker output and earnings increase for all workers in treatment firms, not only those who changed their arrival times as a result of the incentive, suggesting the presence of positive spillovers in treatment firms. Weekly firm output increases by 29%, and revenues increase by 11%. By reducing uncertainty in labor supply to the firm, incentives facilitate better execution of orders, which helps firms sustain buyer relations. I also find evidence for better time and task allocation among late workers in treatment firms, resulting in a 9% increase in output per hour for late workers.

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1 Introduction

As countries develop, they undergo structural changes in the organization of labor – in particular, a shift from self-employment to wage-jobs in firms (Bandiera et al., 2022). However, the take-up of wage jobs in many low-income contexts still remains low. Among those who do take-up wage-jobs, rates of separation and absenteeism remain high, suggesting that the type of wage-jobs offered in such contexts may not match workers' job preferences (Breza and Kaur, 2025). Understanding frictions to wage-employment is thus a crucial topic of research in developing countries. Recent work attributes these low rates of wage-employment to workers' preferences for flexibility (Cefala et al., 2024; Oh et al., 2024).[1] While flexibility is an important job attribute for workers – providing flexibility may be costly for firms, especially in settings where the production process requires a high degree of coordination among workers (assembly line manufacturing). This creates a tension between worker and firm preferences, and may have broader implications for firm growth and economic development if coordination costs increase in scale (Lucas, 1978).

In this paper, I investigate this trade-off between firm and worker preferences for flexible work arrangements, focusing on flexibility in arrival times at work as a dimension of flexibility. I study the context of a large apparel manufacturing cluster in Tirupur, India which features many small-medium firms employing 5-15 workers (tailors). Firms are organized as batch processing lines, where each worker (tailor) performs a distinct operation on the garment in batches of 80-100 pieces. The status quo work arrangement in this setting reflects the trade-off between firm and worker preferences. Tailors are hired on piece rate contracts and organize as independent contractors in firms. This enables them to choose their work timings, instead of being required to follow uniform work shifts like their counterparts working in large/export firms.[2] Effectively, this means that workers do not coordinate their work schedules – in data from 96 firms and 600 workers, the difference in arrival times from firm's stated start time ranges from 3 hours early to 5 hours late, with a 90-10 gap of almost 2 hours (see Figure 1). Moreover, because workers are only incentivized on individual output as opposed to team output (e.g. only 100 sleeves + 100 collars versus 100 pieces of the completed garment), workers' individual choice of work schedules may create a coordination externality on both other workers and consequently the firm owner.

On the other hand, the production technology implies that worker coordination may improve production. *Because* operations need to be performed sequentially to produce the final output, the lack of coordination in worker's arrival times can either (a) limit worker's ability to specialize on their expert task because they need to fill in for a late worker who usually performs the preceding task, or (b) increase idle time if they need to wait for the preceding task to be completed.

[1] For example, Cefala et al. (2024) document flexible work schedules as the second most important reason (after earnings per day) for choosing a casual job over a regular job.

[2] In a separate survey of tailors and firm owners in Tirupur, I document large and systematic differences in pay contracts and work arrangements by firm size – large firms are more likely to hire workers on shift-rate contracts, and enforce strict work timings for workers (see Table A10).

Thus, such flexible work arrangements may slow the overall speed of production, and lower productivity by limiting specialization. This is consistent with owner's beliefs about how flexibility affects production, and why imposing fixed work schedules may increase production: at baseline, owners were asked what type of measures related to their workers would help increase production (un-prompted). Nearly 90% of owners reported that they would like their workers to start work at the same time. This far outweighs other measures – 38% of owners would like workers to take fewer/shorter breaks, 25% would like workers to take fewer un-announced leaves and and 17% to not bring their children to work (Figure 2a). When asked why coordinated timings would help increase production, 66% of owners suggest a reduction in idle time, and 51% suggest increased task specialization as possible mechanisms (Figure 2b). Additionally, 40% owners also allude to uncertainty about whether and when workers will arrive at the firm as another mechanisms. Coordinated timings can help owners plan and deliver orders on time, which helps them receive more orders in the future.

I conduct an experiment with 96 firms and ≈ 600 workers to test whether flexibility in work schedules affects firm productivity. I incentivize workers in treatment firms to follow fixed work start times for a period of 4-weeks. Treatment incentives are designed as a daily bonus of Rs. 100 (20% of average daily earnings), paid conditional on arriving at the firm by the owners' work start time (hereafter, preferred start time), elicited at the time of baseline. Thus, the incentive encourages both attendance and before-time arrival. Workers in control firms receive unconditional transfers during the treatment incentive period, matched to a randomly chosen treatment firm each week to address concerns about income effects on worker output.

I collect daily data on worker attendance, arrival times, breaks taken, and production starting 2-weeks before the incentives, and the 4-week incentive period. I define a worker as compliant with the treatment if they are present *and* arrived on or before the preferred start time. First, I estimate the effect of the incentives on worker attendance and compliance. I interpret the treatment effect on worker compliance as an upper-bound estimate of workers' willingness to pay for workplace flexibility. Second, I study the effect of improved compliance on firm and worker output and (other economic measures). Finally, I empirically test three hypothesized channels of increased firm output: (1) an improvement in the allocation of tasks to workers – either through a decrease in multitasking, or better alignment of tasks to workers' expertise, (2) spillovers along the production line to downstream workers, and (3) an increase in firms' capacity.[3]

I find that treatment significantly increases the probability that workers arrive at work at or before owners' preferred start times as elicited at baseline. This effect is large – the fraction of worker days that workers are present and arrive before the preferred time increases by 18 percentage points, a 43% increase relative to a control mean of 41% before the incentives were active. I find no effect on worker attendance, suggesting that the effect on worker compliance is only driven by changes in workers' arrival times. The effect on compliance is almost entirely driven by workers who

[3] This version of the draft provides suggestive evidence for a subset of these mechanisms.

consistently arrived after the preferred start time at baseline (hereafter, "late" workers), and female workers, who are more likely than male workers to be late at baseline. Workers in treatment firms work on average 16 minutes longer than control firm workers, but this effect is small in magnitude and not statistically significant for the average worker. Similarly, there is no change in the number and total duration of breaks, or the probability of leaving the firm early to compensate for earlier arrival times. Overall, these changes in worker compliance reduce dispersion in worker arrival times at the firm level - total dispersion in worker arrival times, measured as the difference in arrival times between the earliest and the last worker, decreases by 22 minutes (17% of the control mean). The deviation from preferred start time for the 90th percentile worker – which indicates the 2nd to last worker to arrive given firms have an average of 6 tailors – decreases by 14 minutes (29% of the control mean). Thus, incentives are successful in improving coordination in worker arrival times in treatment firms.

Next, I estimate the effect of incentives on overall firm output, measured as the total number of finished pieces produced by all workers in the firm. I find a statistically significant, 0.19 standard deviation increase in weekly firm output – 14% increase relative to the control mean of 2,300 and standard deviation of 1,800. This is equivalent to an increase in total weekly output by $\approx$ 300 pieces. Weekly firm revenues increase by INR 9,300 (31% relative to the control mean), while total labor costs, *including* the incentive paid to treatment workers, increases by INR 8,000 (56% relative to the control mean). Using self-reported monthly non-labor costs incurred by firms, I estimate that treatment firms incurred INR 5,300 lower monthly non-labor costs, or INR 1,300 lower weekly labor costs. A simple profit accounting suggests that weekly firm profits, including incentive payments made to workers, increased by $\approx$ INR 2,600.

To unpack what drives the increase in total firm output, I estimate the intent-to-treat effects of the incentive on *worker* output and earnings. Worker output is the sum of total pieces produced by a worker, across all tasks/products, regardless of whether the garment has been fully stitched. Total worker earnings are calculated as the sum of earnings for each task performed by the worker and summed across all tasks. I find that weekly worker output increases by 0.27 standard deviations (24% relative to the control mean, equivalent to 900 more pieces per week). Importantly, unlike the results on labor supply, increases in weekly worker output and earnings are realized by all workers, and not only driven by female or late workers, suggesting the presence of positive spillovers within the firm. Weekly worker earnings (*excluding* the incentive amount) increase by INR 354 (10% relative to the control mean). This suggests that in the absence of incentives, workers were willing to forgo at least INR 354 in weekly earnings for flexible arrival times. This effect is more pronounced among female workers, and is driven entirely by "late" workers. Including incentive payments, all workers in treatment firms earn INR 861 more per week (25% of the control mean).

Finally, I find that treatment firms receive 0.25 SD larger quantity of new orders during the incentive weeks, equivalent to a 20% increase in new orders received relative to control firms. This is consistent with firm owners' reports at endline that they actively tried to seek new, more profitable orders from both existing and new buyers. Anecdotally, treatment firms owners are 26 percentage points (96%) more likely to agree that total production at their firm increased during the incentive weeks, and are twice as likely to attribute this increase to workers arriving on time.

This paper makes three contributions to the literature. The findings of this paper highlight an important, yet under-explored relationship between the provision of flexible work arrangements, and labor supply preferences (Mas and Pallais, 2020). I experimentally estimate workers' demand for flexible work arrangements and highlight the trade-off between worker preferences for flexibility and firm output. Workers' demand for flexibility has been documented across several different contexts (Mas and Pallais, 2017; Ho et al., 2024; Baysan et al., 2024; Wiswall and Zafar, 2018). This paper's main contribution is to link worker's WTP for flexibility with firm outcomes. I show that while workers individually prefer flexible work arrangements (as demonstrated by (a) the status quo dispersion in arrival times and (b) heterogeneity in response to the incentives), on average, all workers in the firm experience an increase in output and earnings when they arrive by the preferred start time. These findings are consistent with the idea that despite fixed-schedules being a dis-amenity for workers individually, there are potential gains to themselves *and* the firm.

This paper also speaks to a large literature on barriers to firm growth in developing countries. I build on previous work documenting search and hiring frictions in low-income countries (De Mel et al., 2019; Hardy and McCasland, 2023; Bassi and Nansamba, 2022; Caria and Falco, 2024; Carranza et al., 2020) by documenting that even conditional on forming matches, firms may face barriers to coordinating workers. This could have wider economic consequences on firms' ability to scale their workforce as coordination becomes costlier with scale (Lucas, 1978). My results suggest that improving coordination in worker arrival times at the firm do increase firm output and revenues, suggesting that frictions to coordinating workers can pose a significant barrier to firm growth in developing countries.

I contribute to the growing literature on the internal organization of small, informal firms in developing countries, building on recent work such as Bassi et al. (2022, 2023). I document worker preferences as the source of coordination costs for SMEs in developing countries, and how they affect firms' ability to specialize.

The remainder of the paper is organized as follows: section 2 describes the study context and sample firms, section 3 describes the experimental design, implementation and data collection, and section 4 lays out the empirical strategy. section 5 presents the main results and section 6 concludes.

2 Study Context and Sample

The context of my study is Tiruppur district in the state of Tamil Nadu, India. This region is part of a major garment manufacturing hub in India, catering to both export and domestic markets since the 1970s. The production process in Tiruppur is highly decentralized and carried out by sub-contracted units, called job-work units – these units will form my study sample. Job work units are *small* factories, employing about 10-30 workers, at least 50% of whom are women.[4]

Firms are organized as batch processing lines, where each worker performs a distinct operation on the garment in batches of 80-100 pieces. Operations need to be performed sequentially in order to produce a finish piece of garment. Firms typically specialize in broad product categories - such as t-shirts, bottomwear, innerwear, or kids wear. The average weekly order size in these firms is about 4,000 pieces, varying with product type.

The study takes place in 4 neighborhoods located in Tirupur North sub-district, Tirupur, Tamil Nadu. These neighborhoods were identified as high density clusters of small-medium job-work units during prior scoping work and area mapping exercises. I used the following criteria to screen firms to be part of the sample:

1. Has a stitching section
2. Employs at least 2 tailors who are not related to the firm owner or who work for pay (piece rate contract)
3. Must deliver finished products back to the buyer/retailer
4. Must receive cut-pieces of cloth to be stitched (ie does not only produce own designs)

All firms that satisfy these criteria were eligible for baseline. In case eligible firms were located adjacent to each other, I approached alternate firms in order to minimize spillovers between treatment and control firms located next to each other.

3 Experiment Design

Firms are randomly assigned to one of two treatment groups - (T) Treatment (n = 49), and (C) Control (n = 47). All workers in treatment firms were eligible to receive a daily bonus of Rs. 100-150 (equivalent to 20% of their *average* daily earnings) if they arrived at the firm and started work by the company's preferred start time. Thus, the incentive encourages both on-time arrival and attendance. The total incentive amount was paid to workers as part of their weekly earnings.[5] The unconditional incentive amount for control group firms was matched to the total realized payment made to a randomly selected treatment firm, capped at Rs. 300-450 per week (half of the maximum incentive amount in treatment firms). These treatment-control firm (randomized) pairings were created separately each week. Figure 4a shows the experiment design.

[4]Note that these firms are of course larger than the typical small developing country firm; however they employ a substantial fraction of the workforce within the textile industry in Tiruppur district - as of 2013, the share of total employment in textiles in 10-30 worker firms ranged between 20-30% in rural and urban areas in Tiruppur district (source: Economic Census, 2013-14). Such organization structures are common in many industrial clusters around the world – see Nadvi and Schmitz (1994); Banerjee and Munshi (2004); Chaudhry and Woodruff (2013); Atkin et al. (2017).

[5]All workers who were working full-time at the firm at any point during the study were eligible to receive incentives. Workers' full-time status was determined by the owner and verified by the research team using daily attendance data.

3.1 Implementation

The experiment was implemented in three phases - Phase 1 was implemented with 17 firms (9 treatment, 8 control) from November 2024 to January 2025. Phase 2 was implemented with 43 firms (22 treatment, 21 control) from March 2025 to May 2025. Phase 3 was implemented with 37 firms (19 treatment, 18 control) from August 2025 to October 2025. Figure 4b shows the study timeline. The daily treatment incentive amount in Phase 1 and Phase 2 was set to INR 100, and was increased to INR 150 in Phase 3.

Randomization for Phase 1 was conducted by generating a random order of treatment and control assignments, which firms were matched to in the order in which their baseline survey was completed. Randomization for Phases 2 and 3 was stratified on the degree of dispersion in worker arrival times around the owner's preferred work start time, using worker's usual arrival times as reported by the owner at baseline. All specifications reported in the paper control for phase by strata fixed effects to account for the method of randomization in each phase (Bruhn and McKenzie, 2009).

3.2 Data and Summary Statistics

Baseline. I conduct a baseline survey with owners to capture owner and firm characteristics, and a complete worker roster, which is followed by a baseline survey of all workers at the firm. Randomization and treatment announcement was done after owner and worker baseline surveys.

Baseline firm and worker characteristics, and balance by treatment arms are reported in Table 1 and Table 2 respectively. On average, firms hire 6 tailors (excluding owners/co-owners), and have been operating for 6-7 years. Firms produce 4,000 pieces of output each week, and earned a profit of INR 30,000 in the last 30 days (USD 340).

Owners directly supervise workers (that is, without a manager/contractor) in almost all firms – owners also perform stitching work in 87% of firms, and work as regular tailors in about 43% of firms. About 85% owners are male, 57% have completed secondary school and have 19 years of experience in the garment industry.

Workers are on average 37 years old (similar to owners), 44% are male and most workers are married (94%) and have at least one child younger than 14 years of age (85%). 87%add workers live within a 15 minute commute to the firm, have worked at the current firm for 2.5 years and have 15 years of experience in the garment industry. Both firm and worker characteristics are balanced across treatment and control firms (Table 2).

Order, Task and Piece Rates Data: At baseline, I also record a roster of all distinct styles that the firm is currently producing. For each style, I record the piece rate that the owner receives for a finished stitched piece and its task structure – a list of all tasks which they need to be performed to produce the finished product, the order in which they are performed, and task specific piece rates paid to workers. This roster is updated throughout the 6-week study period to incorporate new styles that the firm is producing.

Daily Production and Attendance Data: Starting 1-2 weeks after baseline, I collect data *daily* for a period of 6-weeks, out of which workers are paid incentives for 4-weeks.[6] Each day, I record worker's attendance, arrival time, break times during the day, and daily output (by product-task). Arrival times and attendance are recorded by surveyor observation and are not prone to mis-reporting in order to game the incentive; remaining data is self-reported by workers. Daily output is recorded for all production workers, including firm owner(s), helpers, and temporary workers where applicable. I use daily output data from all production workers to construct a measure of total firm production. In addition to tailoring/production output, I also record whether the worker did any other work at the firm, such as helper work, receiving or delivering raw material/finished pieces, machine repair, purchasing inputs, etc.

Table A1 summarizes key labor supply and production outcomes during the pre-incentive weeks, and differences across two worker types – gender, and whether the worker consistently arrived after the preferred start time during the pre-incentives weeks (hereafter, referred to as "late" versus "early" workers). Column (1) reports summary statistics for the overall worker sample; columns (2)-(3) report summary statistics for female and make workers, and columns (4)-(6) report summary statistics for early, "late", and "very late" workers. "Late" and "very late" are indicators for workers whose average pre-incentive arrival times were between 0 and 30 minutes, and above 30 minutes of the preferred start time, respectively.

Female and male workers have similar demographics – the only exceptions being that women are more likely to be married, live closer to the firm, and have 5 years less experience in the industry than men. On the other hand, late and very late workers are more likely to be female, married, and have more children below 14 years of age. There are no differences in commute times between early, late and very late workers, suggesting that arrival times observed in status quo are not driven by commuting differences, but are likely driven by time commitments in the household before the start of the work day.

Pre-incentive attendance rates are similar across men and women, however, there are several notable gender differences in terms of arrival times and compliance: women arrived 21 minutes after the preferred start time on average, while men arrive 11 minutes before the start time – consequently, women were more likely than men to be "late" and "very late", and 23 percentage points less likely to comply with the preferred start-time - one-third of men's average compliance rate. On the other hand, attendance rates are the highest among late workers at 83%, 10 percentage

[6]In Phase 1, I collected data 1 week pre-incentives and 1 week post-incentives. In Phase 2, I collected 2 weeks of pre-incentives data only, dropping the post-incentive data collection. In Phase 3, I collected 2 weeks of pre-incentives data, 4 week of incentives data and 1 week of post-incentives data.

points higher than attendance rates among early and very late workers.[7] More surprisingly, only 65% of early workers, 45% of late workers, and 10% of very late workers complied with owners' preferred start times before the incentives – suggesting that (a) early workers are not the most reliable workers, and (b) late workers do comply almost half of the time. Given these patterns, I expect the incentive to encourage higher attendance rates among early workers, and to encourage both attendance and on-time arrival among late and very late workers. These patterns also suggest a possible trade-off between regular attendance and on-time arrival. Finally, I observe that women seem to compensate for their late arrival by taking fewer number of breaks during the day, so that average daily hours are only 40 minutes (0.6 hours) lower than men. Late workers work 40 minutes more than both early and very late workers, with no striking differences in the number or duration of breaks taken.

In terms of production outcomes, workers perform 1-2 tasks per day on average, and work on non-tailoring tasks 12% of the time. The average piece rate of tasks performed in a day is 80 paise, with some notable differences by gender – women work on tasks with 30 paise lower average piece rates than men; and baseline lateness – average piece rates decline monotonically with baseline lateness. All workers produce 100 pieces per hours, resulting in small daily earnings differences – women earn Rs. 150 less per day than men, and very late workers earn Rs. 200 less per day than early workers. These differences in piece rates and earnings partly stem from the type of tasks performed – men, and early workers are most likely to perform Complex Singer tasks before the incentives. These tasks are slower and have higher piece rates than the average task.[8]

Orders Data: At the end of each week, I record all orders received and delivered by the firm in the past week. This allows me to track new orders received by firms, defined by new batches of cut-material received, and order completion timelines. Order completion timelines based on owners' reported planned versus actual completion dates per order, allowing me to construct a measure of production delays.

Endline: Finally, I conduct endline surveys during the last week of the study to capture firm owner's and worker's perceptions of the incentives, preferences for flexible work arrangements, changes in firm operations during the study and time-use during and outside of work (relative to the pre-study period). These questions allow me to examine (1) the nature of workers' commitments/restraints that drive their preference for flexibility, (2) how compliance with the incentive impacted these commitments/restraints, and (3) whether compliance shifted workers' preferences

[7] This implies an overall average absenteeism rate of 24%, and 17% among late workers. This range is similar to absenteeism rates documented in other studies. Notably, absenteeism rates among late workers are similar to those reported in large, export factories – 11% in Tanzania (Ho and Huang, 2024), and 15% in India (Adhvaryu et al., 2022). The average absenteeism is similar to those observed in casual labor markets, which are closer to the labor market in this study (25% in rural casual labor markets in Burundi (Cefala et al., 2024), and 20-30% in urban casual labor markets in India (Cefala et al., 2024).

[8] About 50% of all singer tasks are complex tasks, and output per hour on Complex Singer tasks is half of that on other tasks.

for flexible (versus strict) work arrangements.

4 Empirical Strategy

I estimate the effect of incentives on worker level outcomes using the following specification:

$$Y_{i,f,t} = \alpha + \boldsymbol{\beta} T_{i,f} \times \mathbb{I}\{\text{During}\}_t + \lambda_{i,f} + \mu_w + \mu_{dow} + \varepsilon_{i,f,t} \tag{1}$$

where $Y_{i,f,t}$ is the outcome of interest for worker i in firm f at time t, and time is either day or week, depending on the outcome variable. $T_{i,f}$ is an indicator that equals 1 if a firm f is assigned to treatment (fixed over time), $\mathbb{I}\{\text{During}\}_t$ is an indicator variable that takes the value 1 if the study week is an incentive week, and $\lambda_{i,f}$ denotes worker fixed effects. μ_w are study-week fixed effects, [9] and μ_{dow} are day-of-week fixed effects (included only in day-level specifications). $\boldsymbol{\beta}$ is the coefficient of interest, and standard errors $\varepsilon_{i,f,t}$ are clustered at the firm level.

I use a similar specification for firm-level outcomes:

$$Y_{f,t} = \alpha + \boldsymbol{\beta} T_f \times \mathbb{I}\{\text{During}\}_t + \lambda_f + \mu_w + \mu_{dow} + \varepsilon_{f,t} \tag{2}$$

where indices and variables are defined as above, and standard errors $\varepsilon_{f,t}$ are clustered at the firm level. Some firm outcomes are only measured at endline, in which case I will run Equation 2 without the t subscript and without firm and study-week fixed effects. Day-of-week fixed effects are only included in firm-day level regressions.

5 Results

5.1 Labor Supply

First, I test the effect of the incentives on worker attendance, compliance with work start times and daily work hours, reported in Table 3. Attendance is an indicator for whether the worker was present at the firm that day. Compliance is defined as an indicator for being present at the firm, *and* arriving by the owner's preferred start time.[10] Daily work hours are defined as the total work duration net of the total duration of breaks taken during the day. I find no effect on attendance (column (1)),

[9] Study weeks take the value 1 to 4 for the 4 incentive weeks, 0 for the week prior to incentives. Weeks 0-4 are common across all phases. The first pre-incentive week in Phases 2 and 3 is coded week -1, and the final, post-incentive week in Phases 1 and 3 is coded week 5. In the current analysis, I drop the post-incentives week (week 5).

[10] Attendance and compliance are both verified by enumerators.

and an 18 percentage point increase in compliance (column (2)) in treatment firms during the incentives weeks. This represents a 43% increase in compliance relative to the pre-incentive control mean (41 percentage points), and is driven by workers arriving on time, rather than a change in worker attendance. Figure 6 plots the distribution of worker arrival times (conditional on being present) in 15-minute intervals around the preferred start time for treatment (solid lines) and control (dashed lines) firms, visualizing the treatment effect on compliance. Incentives increase the fraction of workers that have arrived within 15 minutes of the preferred start time by 20 percentage points, over a base of 60% in control firms. Given the average size of 6 tailors per firm and 71% attendance rate (4 out of 6 workers present on average), this means that treatment increases number of tailors at the firm at the firm within the first 15 minutes by 1 additional worker.

Column (3) of Table 3 reports the treatment effect on daily work hours. Daily work hours is defined as the total hours worked in a day, excluding the total time spent on breaks. Breaks include any time taken for lunch, tea, or other reasons for which the worker left the firm.[11] Hours are coded 0 on days that a worker was absent. Workers in treatment firms worked an additional 0.27 hours (16 minutes) relative to workers in control firms, but this effect is not significant at conventional levels.

Next, I examine whether these effects vary by gender, and baseline lateness in Table 4. Odd (even) columns report the total effect for the omitted group – males ("early" workers) and differential effects for females ("late" and "very late" workers). As shown in columns (1) and (2), treatment effects on attendance for women and "late" workers are no different than their respective base categories, consistent with the result above that incentives did not change worker attendance in treatment firms. Surprisingly, attendance among early workers does not improve, given that these workers were effectively only incentivized to be present at the firm without changing their arrival time.[12] On the other hand, compliance is driven entirely by workers who were consistently late at baseline, and female workers. Female workers are 13 percentage points more likely than male workers to comply (column 3); "late" and "very late" workers are 21 and 35 percentage points respectively more likely than "early" workers to comply (column 4). This is expected, since the incentives were more binding on workers who consistently arrived after the preferred start-time, which was more common among women. Indeed, the increase in work hours is larger for women (0.44 hours, 26 mins) and late workers (0.34 hours/20 mins and 0.73 hours/44 mins respectively), reported in columns (5) and (6) of Table 4, but not statistically significant at conventional levels.

The magnitude of the increase in work hours – 26 minutes for women and 20-44 minutes for

[11] Workers rarely do other non-tailoring tasks for the firm during breaks (5% of worker-days).

[12] 37% of worker absences are because of social functions, followed by 19% because there was no work at the firm, 18% because either the worker or a family member was sick, and 10% because of local temple/religious festivals. A majority of these reasons relate to importance of social/kinship relations, as emphasized in Breza and Kaur (2025), but it is worth noting a large fraction of absenteeism attributable to low labor demand. This matches the rate of all-day firm closures recorded in the data - 10% of firms were closed for the entire day. 40% of closures were because the firm did not have work, followed by 36% because of local temple festivals or the firm owner's personal commitments, and 10% because of power cuts.

"late" and "very late" workers is largely driven by these subgroups of workers arriving earlier than their pre-incentive start times, as opposed to taking longer breaks during the workday or leaving the firm earlier. Table A2 reports the treatment effect on deviation from the preferred start time. On average, workers in treatment firms arrived 11 minutes earlier relative to control workers. This effect is driven by both male and female workers arriving 7 and 14 minutes earlier respectively (column (2)), and "late" and "very late" workers (column (3)) arriving 12 minutes and 27 minutes earlier relative to control workers. There is some evidence that "very late" workers compensate for arriving earlier by taking one additional break during the day, resulting in an 8 minute increase in the total duration of breaks taken during the day (pval = 0.10), as reported in columns (3) and (6) of Table **??**. Finally, Table A3 suggests there is no evidence that workers leave the firm before 6.30[13] (columns (1)-(3)), before the firm's usual closing time (columns (4)-(6)), or before their own usual exit time (columns (7)-(9)).

Finally, column (4) of Table 3 reports treatment effects for within worker standard deviation in arrival times within a week. This measure reflects consistency in worker arrival times. The average workers' SD in arrival times decreases by 7.6 minutes in treatment firms, relative to a control mean of 43 minutes (18% decrease). Consistent with the patterns highlighted above, this decrease is driven primarily by women (11 minute reduction in arrival time SD, 26% relative to control), and very late workers (23 minute reduction in arrival time SD, 54% relative to control).

5.1.1 Does treatment improve worker coordination in arrival times?

Next, I examine how treatment affects worker coordination at the *firm-day level*, conditional on being present. Table A4 reports treatment effects for three measures of dispersion in worker arrival times at the firm-day level: minutes between the 1st and last worker to arrive at the firm (column (1)), the 90-10 gap in worker arrival times (column (2)), and the deviation from preferred start time for the 90th percentile worker (columns (3)). This regression excludes firm-days on which only 1 worker was present at the firm, since the notion of coordination is not meaningful in such cases.

Total dispersion in worker arrival times is 129 minutes on average at baseline – treatment reduces dispersion by 22 minutes, a 17% reduction in dispersion, or improvement in coordination relative to control firms (column (1)). A more conservative measure that excludes extreme outliers - the 90-10 gap in arrival times – also reduces by 9.5 minutes (p-value ¡ 0.1) in Treatment firms, an 11% decrease relative to the control mean of 84 minutes. Finally, I look at treatment effects on deviation from the owner's preferred start time for the 90th percentile worker. Note that given the average size of firms (6 tailors), and assuming 71% attendance, the 90th percentile worker is equivalent to the last or second to last worker to arrive at the firm. This measure also tests whether these effects are driven by workers for whom the incentive was binding. The deviation of the 90th percentile worker reduces by 14 minutes (p-value ¡ 0.01) in treatment firms, relative to the control group mean of 47 minutes pre-incentives. These effects amount to a 29% reduction in dispersion,

[13] the modal evening tea break time after which workers may leave work early

or improvement in coordination as a result of treatment.

5.2 Worker Output and Earnings

Next, I examine the effect of incentives on worker output and earnings. Worker output is measured as the total quantity of pieces stitched by the worker in a week, across all tasks (and products). This measure does not necessarily correspond to firm output (total number of finished pieces), since a worker who stitched 100 collars and 100 sleeves would be considered to have worked on 200 pieces on that day. However, this measure aligns best with workers earnings, which are constructed as the product of the task specific piece rate and quantity produced on that task, summed over all tasks and products worked on in a week. Output and earnings are coded as 0 on days that a worker was absent, before aggregating to the weekly level. Since firms produce different types of products, I normalize weekly worker output to pre-incentive, product level means and standard deviations to account for any level differences in task speed and production quantity across product types. I report effects on worker earnings both excluding and including treatment incentives.

Worker Output Results for weekly worker output are reported in columns (1)-(3) of Table 5. The treatment effect on weekly worker output is 0.27 standard deviations and statistically significant at the 5% level. This amounts to a 24% increase in weekly output relative to the control group mean (column (1) of Table 5). This increase is driven equally by male and female workers (column (2)), and early and late workers (column (3)). In levels, this amounts to 900 additional pieces per week in treatment firms over an average of 3,700 pieces per week in control firms.

Worker Earnings Results for weekly worker earnings *excluding* incentive payments are reported in columns (4)-(6) of Table 5. I report total worker earnings over the week by summing the product of task-specific output and piece rates over the week, assuming that piece rates account for level differences across products. The treatment effect on worker earnings mirrors the results on worker output, albeit are smaller in magnitude and not statistically significant. Weekly earnings increase by INR 354 (10% of the control mean) for the average worker. The effects are larger among female workers (INR 512, 15% of the control mean, p-val = 0.11), but similar across early and "late" workers. This suggests that if workers were to, hypothetically, arrive on time *without* any monetary incentives, they could have earned on average INR 354-512 more per week than their status quo arrival times, implying a minimum willingness to pay of INR 354 for flexible arrival times (INR 512 for females). *Including* the treatment incentive payments, all workers in treatment firms earn INR 861 per week (see Table A5). The increase in earnings including incentives is experienced by all workers (by design of the incentives), but is the largest among female workers (INR 1,003).

Notably, unlike the results on worker compliance, worker output and earnings are not only concentrated among late and very late workers, suggesting the presence of positive spillovers within the firm as a result of treatment. To assess this further, I first investigate effects on worker output per hour.

5.2.1 Worker output per hour

To what extent are the increases in worker output simply mechanical – i.e., resulting from an increase in work hours at the firm, versus reflecting productivity improvements? To investigate this, and to provide further support for the existence of coordination externalities, I estimate the effects of treatment on worker output per hour, reported in Table 6. For the average worker, output per hour increases by 4.5 units per hour, a 5% increase relative to a control mean of 87 units per hour (column (1)). Consistent with the results on total weekly worker output, this increase is realized by both male/female (column (2)), and early/late workers (column (3)). However, "late" workers experience the largest increase in output per hour - 7.62 units (p-value = 0.09), equivalent to 9% of the control mean. This increase is consistent with "late" workers being less likely to perform non-production tasks at the firm when they are present, as shown in column (3) of Table A6. "Late" and female workers are 5 percentage points less likely to perform non-production tasks[14], equivalent to a 33% reduction relative to the control mean of 16 percentage points. Although not statistically significant, this reduction is economically meaningful to the firm for several reasons. First, this indicates that a subset of tailors are able to specialize in tasks they have a comparative advantage in - tailoring. Second, anecdotally, owners report that tailors typically perform helper tasks when orders are urgent (50%), helpers are absent (33%), there is no other work at the firm (30%), or when tailors have idle time (14%). Table A7 shows that there is no change in helper attendance in treatment firms during the incentive weeks. At endline, treatment firm owners are more likely than control owners to report that both tailors and helpers experienced less idle time at the firm during the incentive period (Figure A1). Together these results provide evidence in support of better time allocation among workers in treatment firms.

5.2.2 Other sources of positive spillover effects

Additionally, two other channels may explain the results on total worker output and output per hour. First, by changing worker arrival times (and consequently, both better coordination in arrival times, and a different order in which workers arrive), treatment may have resulted in a change in task allocation between workers in the firm. It is an open question whether the new task allocation is more efficient for workers and firms. I plan to investigate this further by exploring whether new task allocations are better aligned with worker's expertise[15], or increase the degree of task specialization in the firm. Second, the fact that workers for whom incentives were not binding also benefit from incentives suggests the presence of coordination externalities along the production. For example, increased compliance among late and/or female workers increases their own output

[14] Non-production tasks include helper work, and other work outside the firm like receiving new material, making deliveries, purchasing thread, etc. Most non-production tasks are helper tasks.

[15] Every week, I record worker productivity at the task they are performing at that time. Productivity is measured as the number of pieces of that task completed within 5-minutes and is recorded by surveyor observation. I also elicited which tasks workers consider themselves to be an expert at in worker baseline surveys. I plan to use these data to assess whether treatment improves task allocation by making workers more likely to work on their expert tasks.

each day, but may have spillover benefits to downstream workers (workers who perform the task next in the order). The firm may realize increases in production through the production line even without changes in the task allocation, or changes in the degree of task specialization. These results are work-in-progress and will be included in future revisions of the paper.

5.3 Total Firm Output and Revenue

Finally, how do the changes in individual worker output add up to total firm output and revenues? I construct total firm output using daily production data collected from workers. Total firm output is constructed by adding up total quantity produced on all tasks in the production line, across *all* production workers (i.e., including production by tailors, firm owners and managers, helpers and temporary workers). Firm revenues and total labor costs are calculated using the to quantity produced for the last task in the production line, across all production workers, using piece rates earned by owners (owner piece rate), and total piece rates earned by workers (worker piece rates) respectively. Treatment effects are estimated using Equation 2 and reported in Table 7.

Total firm output increase by 0.31 standard deviations (column (1)) – this is equivalent to 6,000 additional pieces stitched at the firm, a 29% increase relative to the control mean and is statistically significant at the 5% level. This is consistent with the increase in weekly worker output. Using the last task measure for firm revenues, column (2) reports that weekly revenues in treatment firms increased by INR 4,075, an 11% increase relative to the control mean (INR 39,000), but this effect is noisy, and is not statistically significant at conventional levels. Total labor costs including the incentives paid to workers (column (4)) increase by INR 3,400 (21% relative to control firms). Columns (5) and (6) report unit value piece rates[16] earned by owners and workers respectively – notably, while owner piece rates do not change substantially, there is a INR 0.64 decrease in worker piece rates (10% of the control mean), suggesting that firms may have shifted towards products with higher wage markdowns during treatment weeks.

Taken together, columns (1)-(4) suggest that improving worker coordination increases firm output, and increases both firm revenues and labor costs (although noisily estimated). To assess the profitability of incentives, I look at owners' self-reported non-labor costs, which were elicited for the past 30 days at endline. Since endline surveys were conducted in the week immediately after incentives ended, the past month covers the entirety of the incentives weeks in both treatment and control firms. Column (7) of Table 7 reports ANCOVA estimates of non-labor costs, as reported at endline, transformed to a weekly measure. Non-labor costs decreased by INR 1,450 in treatment firms (p-value = 0.02), an 18% decrease relative to the control mean of INR 7,890. This reduction is driven by expenses on transportation (for receiving material/delivering finished pieces) and thread/needle expenses, as shown in columns (2) and (3) of Table A8.

[16] Unit value piece rates are the ratio of total revenue/earnings and total pieces stitched in that week.

5.3.1 Reduction in Non-Labor Costs

Why might non-labor costs decrease as a result of treatment? Conversations with owners during fieldwork consistently the importance of coordination in worker timings on effectively planning and executing orders. Almost all firms in this context engage in repeated orders from buyers/clients they have long-term relationships with. An important component of maintaining such relationships, and receiving recurring orders is delivering completed orders on time. I test this directly using the Orders Data collected from firm owners (subsection 3.2).

Table 8 reports treatment effects on order completion delays, defined as the difference between planned and actual days of delivery (bottom coded at 0 so that positive values indicate larger delays). First, I note that the average delay in control firms, pre-incentives is 0.37 days, the probability of a delay of more than 1 day is 12%, more than 3 days is 4% and more than 7 days is 1%. This suggests that delays are less likely in the status quo, re-iterating owner accounts about the importance of completing orders on time. Columns (1)-(2) report effects on delay in days: there is a 0.69 day reduction in delays on orders received by treatment firms, during incentive weeks. This effect almost twice the control mean of 0.37 days, statistically significant at the 10% level, and does not change after controlling for order characteristics, such as the number new materials received on that order scaled by planned completion days (column (2)). The results on probability of delays (columns (3)-(5)) are even more striking: there is a statistically significant reduction in the probability of a >1 day (>3 day) delay by 13 (10) percentage points, 108% (250%) relative to the control mean pre incentives. The effect on more extreme delays (> 7 days) is also large - 5 percentage points over a control mean of 1 percentage point, but is not statistically significant. Thus, treatment firms are better able to fulfill orders received as a result of increased worker coordination.

Do treatment firms receive different types of orders during the incentive weeks? Table 9 reports treatment effects on the total quantity of new orders received, and an indicator for whether the new order is from the firm's "Main Buyer", defined as the buyer with the largest order size during pre-incentive weeks. There is a 0.25 standard deviation increase in the quantity of new orders received per week in treatment firms (column (1)), equivalent to 800 new pieces, or a 20% increase relative to the control mean. While this effect is not statistically significant, the magnitude is large, and consistent with effects on weekly worker and firm output, suggesting that the increase in production is driven by new orders at the firm as opposed to existing inventory.[17] I find a statistically significant increase in probability that a new order is from the firm's "Main" buyer increases in treatment firms by 9 percentage points, an 11% increase relative to the control mean. A comparison of the control mean across pre-incentive and during-incentive weeks suggests that this increase is driven by a reduction in the probability of producing for the main buyer in control firms. This further lends support to the hypothesis that completing orders on time enables firms

[17]This is also promising, since daily production data and orders data were collected independently from workers and owners respectively.

to get additional orders from the same buyer, and possibly helps sustain long term relationships with buyers.

The increase in the probability of producing for the main buyer makes it plausible that the reduction in thread and transportation costs result from efficiency gains from producing for the same client. For instance, buyers may have fixed preferences for certain types of thread, or producing for the same buyer may help firm owners better coordinate transport to/from the buyers' location. Table A9 provides some anecdotal evidence in support for the former: in the endline survey for phase 3 firm owners (N = 34), I asked whether owners purchased new thread and needle inputs, or used existing inventory of the same inputs in the last 30 days. Treatment firm owners are 19 percentage points less likely to have purchased new inputs, and 24 percentage points more likely to have used existing inputs. Although noisy, the effect on "used existing inputs" is double the control mean, and the magnitudes of both effects suggest that treatment owners substituted towards the use of existing inputs, instead of purchasing new inputs (as opposed to both).

5.3.2 Profitability of Incentives

The estimates shown in Table 7 suggest a range of profit estimates from INR 1450 to INR 2450, depending on assumptions about the magnitude of revenue – 20-32% of baseline weekly profits of INR 7,500 (Table 1). Although noisy, these estimates are large and economically meaningful. Importantly, the reduction in non-labor costs at endline suggests some efficiency gains were realized in the production process, consistent with the reduction in order completion times, increase in orders from repeat buyers in treatment firms, and owners' substituting towards the use of existing inputs. Future iterations of the paper will look further into the distribution of these gains to understand whether certain types of firms are more likely than others to benefit from increased worker coordination.

5.4 Why were owners not incentivizing timely arrival before the experiment?

A natural question arises – if incentives were profitable to firms, why did owners not offer such incentives in the status quo? I provide some suggestive reasons based on survey responses and anecdotes from the field.

First, the existing equilibrium in which these firms operate make it hard for a single firm to enforce common work start times without raising wages or incentivizing workers. Firms are located in dense clusters, where over 95% of firms pay workers on piece rate contracts, and 80% owners report being okay with workers arriving 30 or more minutes late. When asked about the consequences of enforcing common work start times without incentives, owners believe that workers may leave the firm/refused to work at the firm (52%), or they may be less willing to work extra hours when there are urgent orders (38%), and that it may harm their relationship with workers (31%). These reasons point to a combination of hiring frictions, idiosyncratic nature of demand

and reputation concerns for owners.

Second, owner's may have incorrect beliefs about workers' response to such incentives, or incorrect beliefs about the profitability of such incentives. In endline surveys, treatment firm owners are more likely to agree that their firm's total production and profits increased during the incentive weeks. This suggests that the experiment itself may have corrected owners' beliefs about the viability of such incentives. 76% of treatment owners reported that they would be willing to continue the incentives if the research team continued to pay for them, indicating that most owners found the incentives were useful. However, only 26% reported that they would be willing to continue the incentives by paying themselves. On average, treatment owners say that they would be willing to contribute INR 30 per day out of INR 100 for such incentives, and 19 out of 40 owners reported 0, indicating a low willingness to pay.

Finally, it may in fact be too costly to incentivize workers in the long-run. While this paper demonstrates the possibility of gains in the short-run, it is not clear whether workers may continue to arrive on time if incentives were to continue indefinitely. The long-run response of workers depends on several other factors. In particular, missing or thin markets for services such as childcare, transportation (public transit, school buses), or the quality of public infrastructure (municipal water supply, regularity of government ration shops) affect workers' time-use outside of work, and are a key determinant of long-run responses. Perhaps low reported willingness to pay of the owners reflects their beliefs about long-run responses from workers.

Another important long-run consideration is contracting frictions. 7 out of 40 treatment firm owners said that they would be willing to contribute the full amount of the incentive that I offered. However, in conversations with owners after the endline survey, many expressed concerned about being able to implement such incentives themselves. Their reasons ranged from contracting frictions to concerns about fairness within the firm. For example, one owner said: "*I'm willing to offer up to Rs. 200, but if workers get used to it, they may start considering it as part of their salary and still not come on time.*" Another said "*workers did not feel upset because the incentive came from a third party. If I implement the same system, workers would perceive it as partiality.*" The existence of contracting frictions in developing countries is well-known, and is key to understanding why owners were not implementing such incentives before the experiment.

6 Conclusion

This paper experimentally tests whether flexibility in work schedules affects firm productivity in the context of a garment manufacturing cluster in India. I document that workers exercise considerable flexibility in when they arrive at work – at baseline, the 90-10 gap in worker arrival times is almost 2 hours. This stands in contrast to firm owners' preference to have workers arrive at a common time, given the production line based organization of firms.

I conduct an experiment with 96 small-medium firms where workers in treatment firms are incentivized to arrive before the owner's preferred start time for a period of four weeks. Incentives increase timely arrival by 40%, resulting in a 35% decrease in dispersion in arrival times at the firm level. This is a meaninful effect given the small size of these firms – incentives increase the fraction of workers who arrive within the first 45 minutes of opening time from 60% (3 out of 5 workers) to 80% (4 out of 5 workers).

As a result of the incentives, average weekly revenues increase by 11% in treatment firms, and the increase in revenues exceeds the increase in labor costs *including* incentives, although these estimates are noisy. There is however a statistically significant and large reduction in weekly *non-labor costs* in treatment firms by 18% relative to control firms. This yields a range of treatment effects on weekly profits from INR 1,450 to INR 2,450, indicating that offering these incentives could be profitable for firms. However, only 26% owners said they would be willing to offer such incentives themselves. On average, owners are willing to pay INR 30 per day for such incentives, less than a third of the incentives that I offered in the experiment, indicating low willingness to pay.

Survey responses and anecdotal evidence suggests several possible constraints explaining why willingness to pay is low, and why owners were not implementing such incentives in the status quo. These reasons range from incorrect information and beliefs to norms, missing markets and contracting frictions that are prevalent in many low-middle income contexts. The latter three may require larger scale policy efforts to shift owners' willingness to change the types of contracts and incentives offered to workers. My results demonstrate the importance of policies aimed at improving supply of and access to childcare, and improving the quality of public services such as transportation and municipal residential services (e.g. water supply) as complementary to industrial policies. This also motivates the need for future research to understand what drives the nature of employment contracts offered by firms in developing countries, and their effects on firm productivity.

Finally, it is also striking that incentives did not change worker attendance, and that take-up does not increase to 100% workers arriving on time conditional on attendance. This is despite the incentives being 20% of average daily earnings, indicating that some workers very highly value flexibility. In other survey work with workers across Tirupur district, I document that the provision of flexibility varies systematically with firm size – small and medium firms are significantly more likely than large firms to offer their workers flexibility in work schedules. Consequently, there is sorting of workers into small ($\leq$ 20 workers) versus large ($>$ 20 workers) firms along gender, martial status and having children – characteristics that positively correlated with demand for flexibility. While workers who work in small firms enjoy flexible work schedules, they do so at the cost of variable pay and lack of access to formal government benefits, such as state provided health insurance, social security, etc. To the extent that demand for flexibility affects workers' labor supply choices, such as whether to be self-employed, work as casual day labor (Oh et al., 2024; Cefala et al., 2024), or work in small firms without formal benefits, this can have important implications for the rate of wage-employment in developing countries. More research is needed to understand the drivers of the demand for flexibility in developing countries, as well as the broader economic implications on the functioning of labor markets and firms in such settings.

References

Adhvaryu, A., N. Kala, and A. Nyshadham (2022). Management and shocks to worker productivity. *Journal of Political Economy 130*(1), 1–47.

Atkin, D., A. Chaudhry, S. Chaudhry, , A. K. Khandelwal, and E. Verhoogen (2017). Organizational barriers to technology adoption: Evidence from soccer-ball producers in pakistan. *The Quarterly Journal of Economics 1101*, 1164.

Bandiera, O., A. Elsayed, A. Heil, and A. Smurra (2022). Economic development and the organisation of labour. *Working Paper*.

Banerjee, A. and K. Munshi (2004). How efficiently is capital allocated? evidence from the knitted garment industry in tirupur. *The Review of Economic Studies 71*(1), 19–42.

Bassi, V., J. H. Lee, A. Peter, T. Porzio, R. Sen, and E. Tugume (2023). Self-employment within the firm. Technical report, National Bureau of Economic Research.

Bassi, V., R. Muoio, T. Porzio, R. Sen, and E. Tugume (2022). Achieving scale collectively. *Econometrica 90*(6), 2937–2978.

Bassi, V. and A. Nansamba (2022). Screening and signalling non-cognitive skills: experimental evidence from uganda. *The Economic Journal 132*(642), 471–511.

Baysan, C., M. H. Dar, K. Emerick, Z. Li, and E. Sadoulet (2024). The agricultural wage gap within rural villages. *Journal of Development Economics 168*, 103270.

Breza, E. and S. Kaur (2025). Labor markets in developing countries. *Annual Review of Economics 17*.

Bruhn, M. and D. McKenzie (2009). In pursuit of balance: Randomization in practice in development field experiments. *American economic journal: applied economics 1*(4), 200–232.

Caria, S. A. and P. Falco (2024). Skeptical employers: Experimental evidence on biased beliefs constraining firm growth. *Review of Economics and Statistics 106*(5), 1352–1368.

Carranza, E., R. Garlick, K. Orkin, and N. Rankin (2020). Job search and matching with two-sided limited information.

Cefala, L., S. Kaur, H. Schofield, and Y. Shamdasani (2024). Habit formation in labor supply.

Cefala, L., P. Naso, M. Ndayikeza, and N. Swanson (2024). Under-training by employers in informal labor markets: Evidence from burundi. Technical report, Working Paper.

Chaudhry, T. and C. Woodruff (2013). Incentives and productivity: Work groups vs. production lines. *IGC Working Paper (F-6005-PAK-1)*.

De Mel, S., D. McKenzie, and C. Woodruff (2019). Labor drops: Experimental evidence on the return to additional labor in microenterprises. *American Economic Journal: Applied Economics 11*(1), 202–235.

Hardy, M. and J. McCasland (2023). Are small firms labor constrained? experimental evidence from ghana. *American Economic Journal: Applied Economics 15*(2), 253–284.

Ho, L., S. Jalota, and A. Karandikar (2024). Bringing work home: Flexible work arrangements as gateway jobs for women in west bengal. Technical report, Technical Report.

Ho, Y. and Y. Huang (2024). Selection and sorting when supervisors have discretion: Experimental evidence from. *Working Paper*.

Lucas, R. E. (1978). On the size distribution of business firms. *The Bell Journal of Economics*, 508–523.

Mas, A. and A. Pallais (2017). Valuing alternative work arrangements. *American Economic Review 107*(12), 3722–3759.

Mas, A. and A. Pallais (2020). Alternative work arrangements. *Annual Review of Economics 12*(1), 631–658.

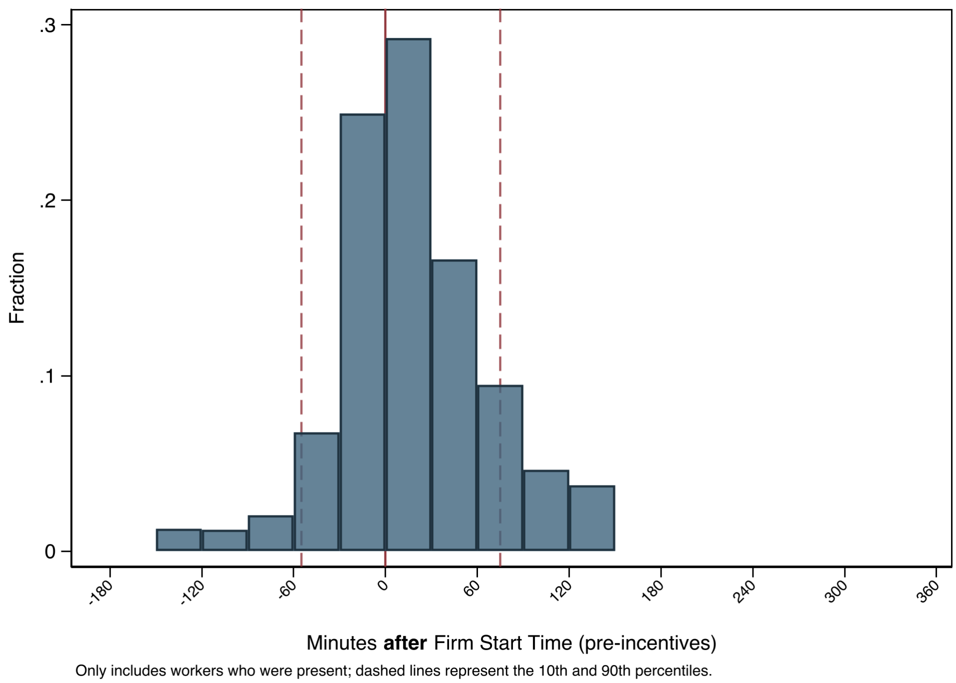
Nadvi, K. and H. Schmitz (1994). *Industrial clusters in less developed countries: review of experiences and research agenda*, Volume 39. Institute of Development Studies Brighton.

Oh, S., Y. Shamdasani, and S. Goraya (2024, September). Worker absences and demand for flexible contracts.

Wiswall, M. and B. Zafar (2018). Preference for the workplace, investment in human capital, and gender. *The Quarterly Journal of Economics 133*(1), 457–507.

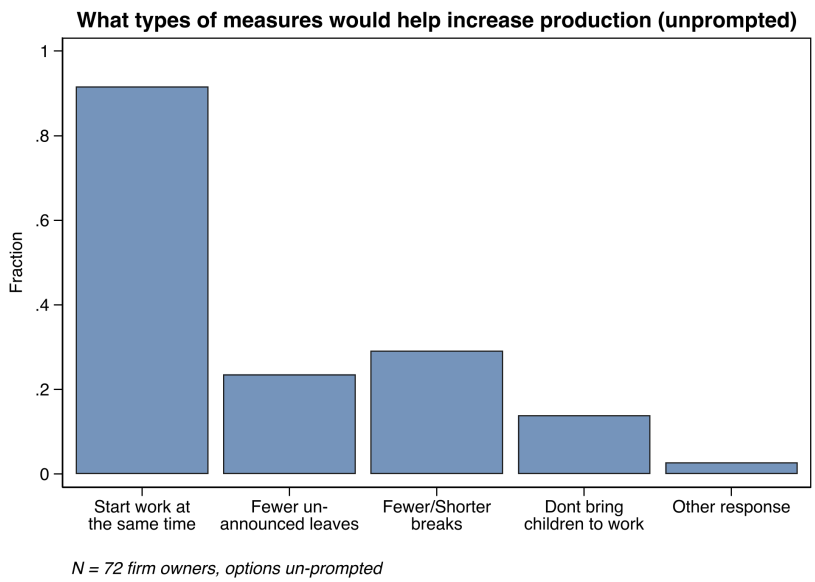
7 Figures

Figure 1: Dispersion in Worker Arrival Times, before Incentives

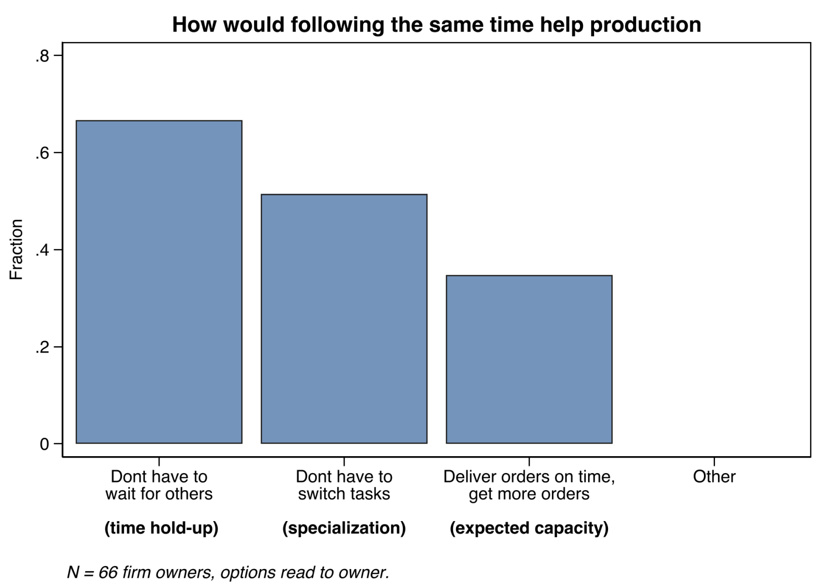


Notes: This figure plots the distribution of the deviation of worker arrival times from the owner's preferred start time. N = 96 firms, $\approx$ 600 workers.

Figure 2: Importance of Common Arrival Times (Baseline)



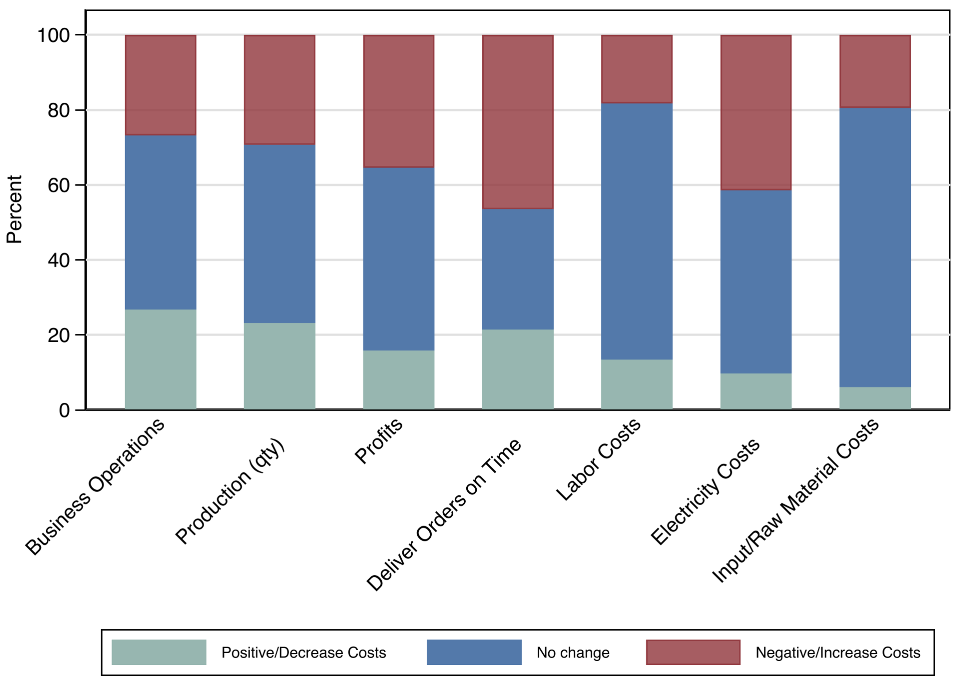
(a) Measures that could help increase production



(b) Why is coordination in worker arrival times important

Notes: Panel A reports owner's responses from the question "*What type of measures would help increase production?*", with options unprompted. N = 71. Panel B reports owner's responses from the question "*If everybody starts work at the same time, how will it improve production?*", with options 1 = Workers don't have to wait for others, 2 = Workers don't have to switch tasks, 3 = Helps me fulfill orders on time and get more orders, and 4 = Others.

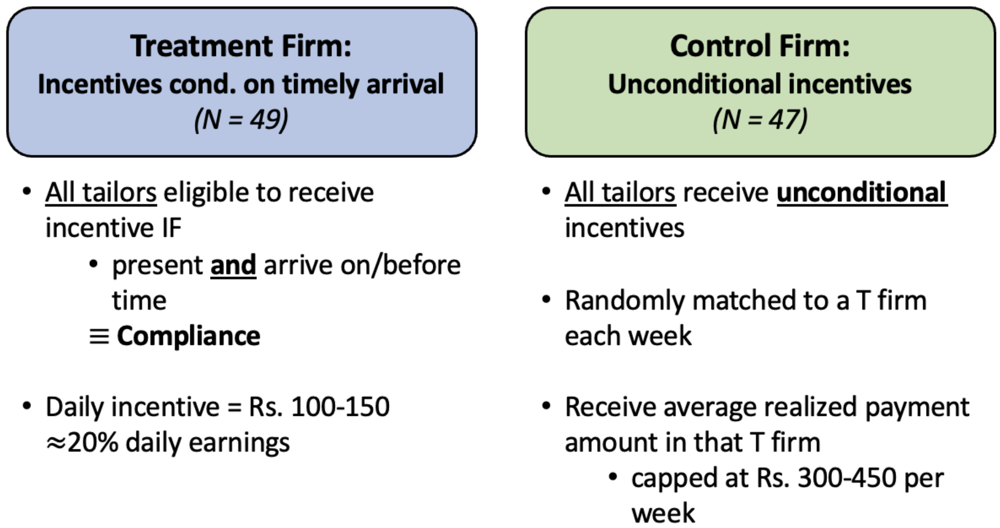
Figure 3: How Providing Flexibility Affects Firm Operations



Notes: This figure reports owner's responses from the question: "*Now I will list 4 different business objectives. For each of these, do you think providing workers flexibility affects this objective positively, negatively, or has no effect?*" This data is from the full sample of 96 firm owners.

Figure 4: Experimental Design and Implementation Timeline

(a) Experiment Design



(b) Study Timeline

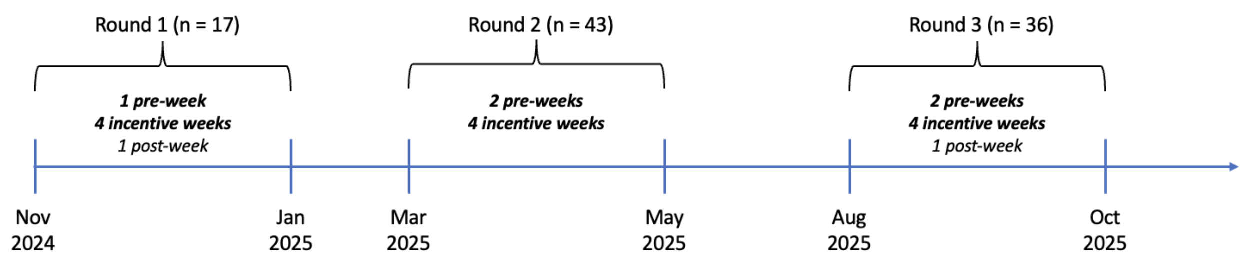


Figure 5: Change in deivation from owner's preferred start time

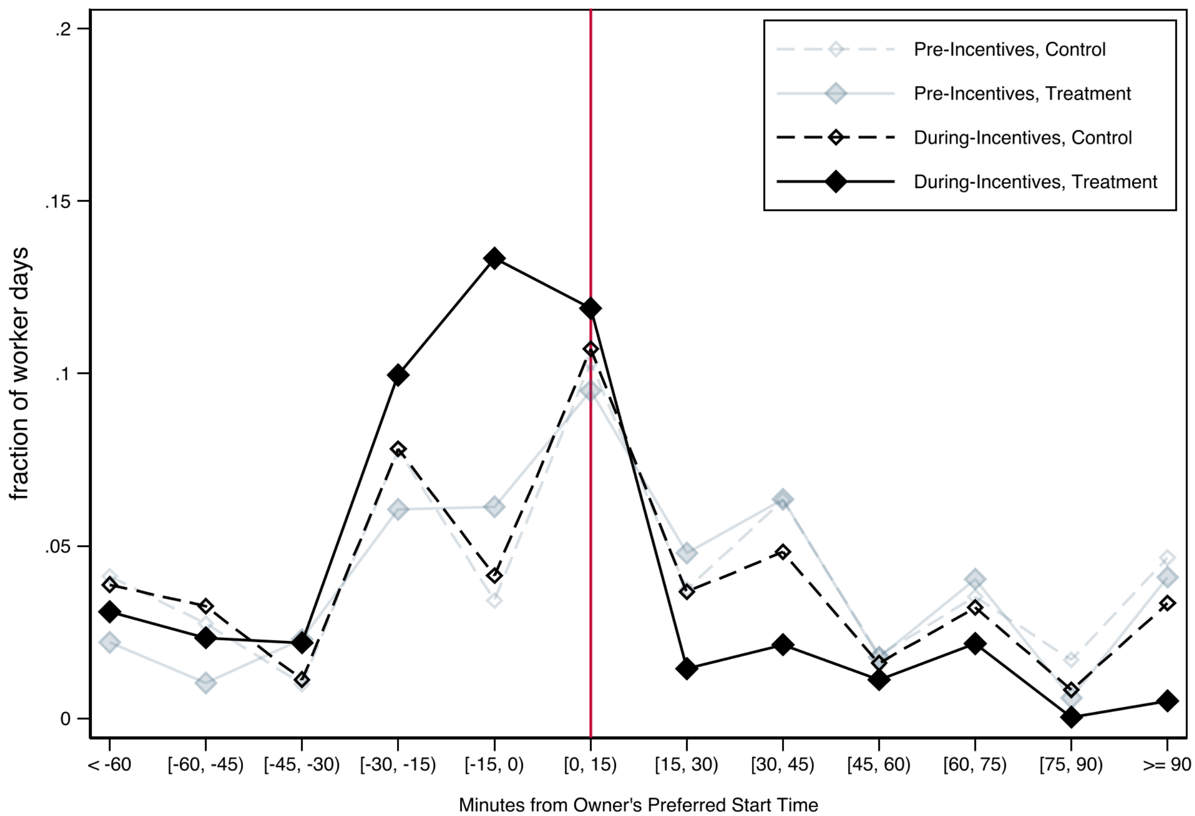
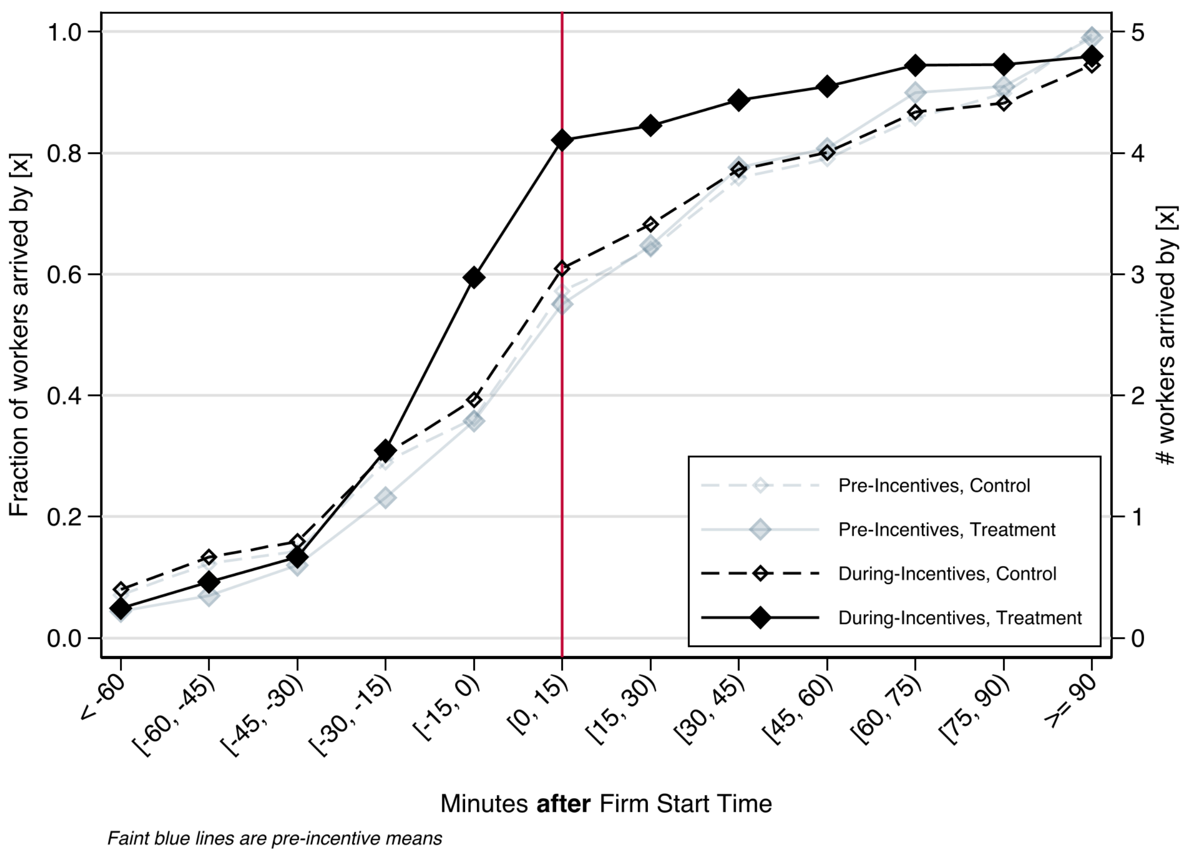


Figure 6: CDF of fraction of workers arriving in 15-minute intervals



8 Tables

Table 1: Balance on Baseline Firm Characteristics

	Control	Treatment	T - C	*p-value*
	Mean/SD	Mean/SD	Difference	
Firm Characteristics				
# Tailors	6.47	6.22	-0.18	0.75
	(2.93)	(2.90)		
# Helpers	1.64	1.29	-0.37	0.12
	(1.15)	(1.15)		
Total # Machines	11.64	11.06	-0.44	0.59
	(4.86)	(3.63)		
Owner directly supervises workers	0.91	0.98	0.06	0.20
	(0.28)	(0.14)		
Firm Age (yrs)	6.68	6.76	0.05	0.97
	(5.29)	(6.18)		
Export orders	0.13	0.16	0.04	0.61
	(0.34)	(0.37)		
Only stitching section	0.57	0.63	0.05	0.59
	(0.50)	(0.49)		
Weekly Output	4075.61	4306.98	286.90	0.59
	(2320.43)	(2876.29)		
Revenues (last 30 days)	134444.19	126900.00	-8054.10	0.70
	(118877.68)	(101095.00)		
Profits (last 30 days)	33962.98	25916.67	-8173.26	0.36
	(58396.33)	(18038.77)		
Total Costs - wage bill (last 30 days)	32478.68	32978.37	842.78	0.83
	(20378.58)	(18684.70)		
Owner Demographics				
Age	39.21	37.80	-1.28	0.44
	(8.36)	(8.77)		
Male	0.85	0.82	-0.03	0.73
	(0.36)	(0.39)		
Completed Secondary School	0.57	0.55	-0.01	0.88
	(0.50)	(0.50)		
Commute Time (mins)	6.32	8.63	2.38	0.15
	(6.66)	(9.06)		
Experience in Industry (yrs)	19.15	18.35	-0.91	0.65
	(10.47)	(9.50)		
Owner does stiching work	0.87	0.88	0.00	1.00
	(0.34)	(0.33)		
Owner stitches as a regular tailor	0.43	0.43	-0.01	0.95
	(0.50)	(0.50)		
Observations	47	49		
Joint F-test p-value				0.475

Notes: This table reports means, mean differences by treatment group and p-values for baseline firm and owner characteristics. Mean differences and p-values are estimated using a regression of each variable on an indicator for treatment with phase x strata fixed effects, with robust standard errors. The joint F-test p-value is from a regression of treatment on all balance variables with strata fixed effects and robust SEs; missing values in weekly output, revenue, and profits are mean-imputed within strata for this test only.

Table 2: Balance on Baseline Worker Characteristics

	Control	Treatment	T - C	*p-value*
	Mean/SD	Mean/SD	Difference	
Age	37.22	36.49	-0.78	0.40
	(8.75)	(9.01)		
Male	0.45	0.43	-0.02	0.73
	(0.50)	(0.50)		
Married	0.94	0.91	-0.04	0.38
	(0.43)	(0.46)		
Has children < 14 yrs	0.89	0.84	-0.06	0.71
	(1.26)	(1.16)		
Has elderly member in HH	0.36	0.40	0.04	0.41
	(0.48)	(0.49)		
Related to owner	0.21	0.21	0.00	0.99
	(0.41)	(0.41)		
Native to TN	0.92	0.95	0.03	0.24
	(0.27)	(0.21)		
Commute < 15 mins	0.87	0.84	-0.03	0.37
	(0.34)	(0.37)		
Completed Secondary School	0.45	0.40	-0.05	0.29
	(0.50)	(0.49)		
Experience in Industry (yrs)	15.52	14.15	-1.43	0.14
	(9.25)	(9.53)		
Tenure at Firm (yrs)	2.57	2.38	-0.17	0.73
	(3.60)	(3.20)		
Observations	304	296		
Joint F-test p-value				0.333

Notes: This table reports means, mean differences by treatment group and p-values for baseline firm and owner characteristics. Mean differences and p-values are estimated using a regression of each variable on an indicator for treatment with phase x strata fixed effects, with standard errors clustered at the firm level. The joint F-test p-value is from a regression of treatment on all balance variables with strata fixed effects and SEs clustered at the firm level.

Table 3: Worker Attendance and Compliance

	Present	**Present and Arrived on Time**	**Daily Work Hours**	**w/n Week SD Arrival Times (mins)**
	(1)	(2)	(3)	(4)
Treatment × During Incentives	0.02	0.18***	0.27	-7.64
	[0.03]	[0.03]	[0.31]	[4.62]
Control Mean (Pre-Incentives)	.71	.41	6.41	42.84
Control Mean (Incentive Weeks)	.68	.43	6.79	34.81
Treatment Effect %	3	43	4	-18
Worker, Week FE	X	X	X	X
Day-of-week FE	X	X	X	
Worker-Days	17705	17703	16633	
Worker-Weeks				2365

Notes: All specifications include worker, day-of-week, phase by week, phase by strata fixed effects. Standard errors are clustered at the firm level. Attendance is defined as an indicator for whether a worker was present at the firm on that day. Compliance is defined as an indicator for whether a worker was present at the firm, *and* arrived by the owner's preferred start time on a particular day. The sample for these regressions excludes days on which a firm was shut, and no production took place.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Heterogeneity in Treatment Effects on Compliance and Attendance

	Attendance		**Compliance**		**Work Hours**		**Arrival Times SD (mins)**	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treatment × During	0.02	0.03	0.10***	0.01	0.05	-0.05	-2.90	-2.64
	[0.03]	[0.04]	[0.04]	[0.04]	[0.32]	[0.41]	[4.86]	[4.34]
Treatment × During × Female	0.01		0.13***		0.39		-8.17	
	[0.04]		[0.04]		[0.32]		[7.31]	
Treatment × During × Late		-0.01		0.21***		0.39		2.11
		[0.05]		[0.05]		[0.47]		[5.29]
Treatment × During × Very Late		0.00		0.35***		0.78		-20.51**
		[0.05]		[0.06]		[0.51]		[9.37]
Control Mean (Pre-Incentives)	.71	.71	.41	.41	6.41	6.41	42.84	42.84
Omitted Group Mean (Pre-Incentives)	.71	.78	.55	.69	6.69	7.33	35.16	36.25
Total Effect (T × During + Inter.)	0.03	0.02, 0.03	0.23	0.22, 0.36	0.44	0.34, 0.73	-11.06	-0.53, -23.14
p-val Total Effect = 0	0.45	0.63, 0.53	0.00	0.00, 0.00	0.20	0.30, 0.12	0.09	0.92, 0.02
Treatment Effect %	4	3, 4	57	54, 88	7	5, 11	-26	-1, -54
Worker, Week FE	X	X	X	X	X	X	X	X
Day-of-week FE	X	X	X	X	X	X		
N Worker-Days	17705	15889	17703	15888	16633	15308		
N Worker-Weeks							2365	2342

Notes: All specifications include worker, day-of-week, phase by week, phase by strata fixed effects, from the specification in Equation 1. Standard errors at clustered at the firm level. Attendance is defined as an indicator for whether a worker was present at the firm on that day. Compliance is defined as an indicator for whether a worker was present at the firm, *and* arrived by the owner's preferred start time on a particular day. The sample for these regressions excludes days on which a firm was shut, and no production took place.
Total Effect reports the lincom of T × During plus the listed interaction (Late, Very Late shown comma-separated). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Weekly Output and Earnings

	Weekly Output (SD)			Weekly Earnings (INR)		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment × During	0.27**	0.25*	0.22	353.88	151.99	346.54
	[0.12]	[0.14]	[0.14]	[317.81]	[362.87]	[395.35]
Treatment × During × Female		0.04			360.23	
		[0.10]			[233.40]	
Treatment × During × Late			0.09			-104.07
			[0.14]			[364.01]
Treatment × During × Very Late			0.07			133.86
			[0.14]			[315.10]
Control Mean (Pre-Incentives)	3765.38	3765.38	3765.38	3496.51	3496.51	3496.51
Control SD (Pre-Incentives)	3384.88	3384.88	3384.88			
Omitted Group Mean (Pre-Incentives)		3621.99	3733.12		4041.7	4045.79
Total Effect (T × During + Inter.)		0.28	0.31, 0.29		512.22	242.47, 480.40
p-val Total Effect = 0		0.02	0.03, 0.03		0.11	0.52, 0.13
Treatment Effect %	24	26	28, 26	10	15	7, 14
N Worker-Weeks	2667	2667	2602	2667	2667	2602

Notes: SEs clustered at firm level. All specifications include worker, phase by week, phase by strata fixed effects. This regression excludes (a) owners/co-owners who also do stitching work at the firm, and (b) days of firm closure if no production took place at the firm. Output is normalized by product relative to pre-incentive week means. Earnings are coded as 0 for absent workers. Output and earnings are winsorized at the 95th percentile. Total Effect reports the lincom of T × During plus the listed interaction (Late, Very Late shown comma-separated).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: Worker Output per hour

	Output per hour | present					
		Daily			Weekly	
	Baseline	Female	Late	Baseline	Female	Late
Treatment × During	4.52	5.78	4.01	5.12	5.73	5.47
	[4.33]	[4.73]	[5.17]	[5.20]	[5.71]	[6.02]
Treatment × During × Female		-2.19			-1.07	
		[2.95]			[3.81]	
Treatment × During × Late			3.61			3.39
			[4.13]			[5.26]
Treatment × During × Very Late			-2.52			-5.10
			[3.96]			[4.86]
Control Mean (Pre-Incentives)	86.78	86.78	86.78	89.2	89.2	89.2
Omitted Group Mean (Pre-Incentives)		79.83	82.94		81.38	84.9
Total Effect (T × During + Inter.)		3.59	7.62, 1.49		4.66	8.86, 0.37
p-val Total Effect = 0		0.42	0.09, 0.76		0.39	0.14, 0.95
Treatment Effect %	5	4	9, 2	6	5	10, 0
N Worker-Days/Weeks	11787	11787	11676	2402	2402	2378

Notes: Present-sample regressions on regular-position workers (worker$_{p}osition$=3)$; cols1 - -3daily, cols4 - -6weekly. All specifications include worker, phase by week, phase by strata fixed effects; SEs (in brackets) clustered at the firm level. Late flags workers l$
During plus the listed interaction (Late, Very Late shown comma-separated). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Treatment Effects on Weekly Firm Output

	Production Data						Endline
			Total Labor Cost (INR)		Piece Rates (INR)		
	Pieces (SD)	**Revenues (INR)**	w/o subsidy	w/ subsidy	Owner	Worker	**Non Lab. Costs**
	All Tasks	*Last Task*	*Last Task*				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treatment × During	0.31**	4075.05	785.63	3384.30	0.18	-0.64	
	[0.15]	[4248.60]	[2098.35]	[2213.35]	[1.06]	[0.74]	
Treatment							-1449.06**
							[602.59]
Control Mean (Pre-Incentives)	22633.35	38622.65	15949.73	15949.73	15.58	7.31	
Control SD (Pre-Incentives)	20714.59						
Control Mean (Endline)							7884.88
p-val (T × D / T)	0.05	0.34	0.71	0.13	0.87	0.39	0.02
Treatment Effect %	29	11	5	21	1	-9	
Firm FE	X	X	X	X	X	X	
Baseline control							X
Observations	529	529	529	529	483	483	90

Notes: SEs in brackets, clustered at the firm level. Data at the firm-week level. Total firm output is calculated using all tasks along the production line. Revenues and labor costs are calculated using the total nuber of finished pieces (calculated using the last task on the production line). Worker payments are computed using the sum of task-specific piece rates for all tasks that constitute the final product. Non labor costs are self-reported at endline (separate data source). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Order Completion Time - Delays

	Delay in days		$\mathbb{I}\{$**Delay** $>\}$		
	Delay in days		**1 day**	**3 days**	**7 days**
	(1)	(2)	(3)	(4)	(5)
Treatment × During	-0.69*	-0.71*	-0.13**	-0.10**	-0.05
	[0.39]	[0.38]	[0.05]	[0.05]	[0.04]
New materials (qty) per day		0.00***			
		[0.00]			
Control Mean (Pre-Incentives)	.37	.37	.12	.04	.01
p-val (T × D)	0.08	0.06	0.01	0.04	0.16
Observations	1080	1080	1080	1080	1080

Notes: Data at order-week level. Standard errors are clustered at the firm level. Delay = difference between planned and actual delivery timelines. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9: New Orders (SD)

	Total Orders (SD)	$\mathbb{I}$\{**Main Buyer**\}
	(1)	(2)
Treatment × During	0.25	0.09*
	[0.19]	[0.05]
Control Mean (During-Incentives)		.61
Control Mean (Pre-Incentives)	4196.55	.78
Control SD (Pre-Incentives)	3244.54	
p-val (T × D)	0.20	0.07
Observations	386	1026

Notes: Standard errors are clustered at the firm level. Quantity is standardized to pre-incentive product level means. Column (1) at firm-pre/during week level; columns (2)-(3) at order level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A Appendix Tables and Figures

Table A1: Worker Labor Supply and Production Outcomes - Pre Incentives

		Avg. Dev. in Arrival Time before incentives					**By Gender**		
	All workers	**≤ 0 (Early)**	**$\in (0,30]$ mins (Late)**		**> 30 mins (Very Late)**		**Female**	**Male**	
	mean	mean	mean	p-val	mean	p-val	mean	mean	p-val
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Attendance, Arrival Times									
Present	0.73	0.78	0.83	0.95	0.75	0.01	0.74	0.72	0.83
Complied w preferred start time	0.41	0.70	0.44	0.00	0.08	0.00	0.31	0.54	0.00
Dev. in Arrival Time (mins)	8.05	-27.44	14.33	0.00	55.04	0.00	25.14	-13.51	0.00
Dev. in Exit Time (mins)	-37.16	-32.79	-36.34	0.59	-44.81	0.23	-41.49	-31.70	0.05
Daily Work Hours	6.61	7.48	7.55	0.20	6.11	0.00	6.39	6.87	0.03
Total # Breaks	1.84	2.10	2.05	0.13	1.74	0.00	1.81	1.89	0.29
Total Break Duration - mins	58.56	65.11	66.83	0.82	55.87	0.00	56.65	60.94	0.14
Production									
Avg Task Piece Rate (INR)	0.92	1.29	0.85	0.08	0.71	0.02	0.72	1.18	0.00
Output per hour	87.86	86.89	92.27	0.85	84.31	0.27	88.64	86.89	0.84
Earnings per hour (INR)	71.62	78.53	69.39	0.38	63.59	0.07	65.06	79.90	0.00
Daily Earnings (INR)	510.80	647.86	558.80	0.47	396.03	0.00	430.90	609.85	0.00
Upstream Task	0.26	0.31	0.32	0.93	0.22	0.04	0.27	0.26	0.83
Downstream Task	0.30	0.36	0.33	0.46	0.24	0.02	0.24	0.37	0.00
Complex Singer Task	0.11	0.19	0.08	0.00	0.06	0.00	0.05	0.19	0.00
Demographics									
Female	0.55	0.32	0.60	0.00	0.84	0.00			
Married	0.83	0.79	0.86	0.06	0.93	0.00	0.90	0.75	0.00
# children < 14 yrs	0.93	0.83	1.03	0.30	0.99	0.76	1.02	0.82	0.25
Commute < 15 mins	0.85	0.81	0.86	0.46	0.88	0.13	0.88	0.81	0.14
Tenure at Firm	2.85	2.74	3.32	0.10	2.79	0.76	2.64	3.12	0.33
Experience in Industry (yrs)	15.19	16.45	14.52	0.02	13.69	0.00	12.80	18.31	0.00
Owner Baseline Ranking									
Punctual	0.16	0.24	0.16	0.07	0.07	0.00	0.13	0.21	0.01
Reliable	0.18	0.19	0.21	0.77	0.14	0.31	0.14	0.23	0.02
Productive	0.17	0.22	0.16	0.18	0.12	0.01	0.15	0.19	0.16
Knowledgable about Prod.	0.16	0.20	0.15	0.33	0.14	0.28	0.12	0.22	0.01
benefit most from intervention	0.17	0.20	0.16	0.57	0.16	0.40	0.16	0.19	0.19
Observations	5096	1980	1399		1361		2819	2277	
Fraction of Workers		0.40	0.30		0.29		0.55	0.45	

Notes: Data at the worker-day level, includes 1-2 weeks of pre-incentives data, excludes days firm was closed. All outcomes coded as 0 if worker was absent. Piece rate and earnings are in INR.

Table A2: Treatment Effects on Deviation from Start Time

	Deviation from Start Time (mins)		
	(1)	(2)	(3)
Treatment × During	-11.01***	-7.44**	0.67
	[2.74]	[2.94]	[2.60]
Treatment × During × Female		-6.19**	
		[2.38]	
Treatment × During × Late			-12.74***
			[2.73]
Treatment × During × Very Late			-27.57***
			[4.50]
Control Mean (Pre-Incentives)	5.88	5.88	5.88
Omitted Group Mean (Pre-Incentives)		-18.31	-30.73
Total Effect (T × During + Inter.)		-13.63	-12.06, -26.90
p-val Total Effect = 0		0.00	0.00, 0.00
Treatment Effect %	-187	-232	-205, -458
Worker, Week FE	X	X	X
Day-of-week FE	X	X	X
N Worker-Days	12477	12477	12327

Notes: All specifications include worker, day-of-week, phase by week, phase by strata fixed effects, from the specification in Equation 1. Standard errors at clustered at the firm level. The sample for these regressions excludes days on which a firm was shut, and no production took place.
Total Effect reports the lincom of T × During plus the listed interaction (Late, Very Late shown comma-separated). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A3: Treatment Effects on Worker Exit Times

	$\mathbb{I}${**Left Before**}						**Minutes after**		
	6.30 pm			**Closing Time**			**Regular Closing Time**		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treatment × During	0.00	0.02	0.04	-0.01	-0.01	-0.02	-7.14	-12.37	-14.42
	[0.02]	[0.03]	[0.04]	[0.03]	[0.04]	[0.04]	[7.39]	[8.42]	[9.11]
Treatment × During × Female		-0.03			0.01			9.06*	
		[0.03]			[0.04]			[5.31]	
Treatment × During × Late			-0.02			0.00			7.34
			[0.04]			[0.05]			[7.90]
Treatment × During × Very Late			-0.07**			0.03			17.61*
			[0.04]			[0.06]			[9.51]
Control Mean (Pre-Incentives)	.13	.13	.13	.57	.57	.57	-38.72	-38.72	-38.72
Omitted Group Mean (Pre-Incentives)		.13	.16		.58	.63		-37.67	-38.38
Total Effect (T × During + Inter.)		-0.01	0.01, -0.04		-0.01	-0.01, 0.02		-3.31	-7.08, 3.19
p-val Total Effect = 0		0.74	0.66, 0.21		0.87	0.77, 0.76		0.65	0.33, 0.74
Treatment Effect %	2	-7	11, -30	-2	-1	-2, 3	18	9	18, -8
Worker, Week FE	X	X	X	X	X	X	X	X	X
Day-of-week FE	X	X	X	X	X	X	X	X	X
N Worker-Days	17705	17705	15889	17705	17705	15889	12279	12279	12131

Notes: All specifications include worker, day-of-week, phase by week, phase by strata fixed effects. Standard errors are clustered at the firm level. Total Effect reports the lincom of T × During plus the listed interaction (Late, Very Late shown comma-separated).

	# Breaks			Total Break Duration (mins)		
Treatment × During	0.16	0.11	0.06	3.73	2.22	1.72
	[0.10]	[0.11]	[0.14]	[3.17]	[3.49]	[4.33]
Treatment × During × Female		0.09			2.65	
		[0.11]			[3.56]	
Treatment × During × Late			0.07			1.17
			[0.17]			[5.29]
Treatment × During × Very Late			0.30*			6.94
			[0.15]			[5.32]
Control Mean (Pre-Incentives)	1.85	1.85	1.85	59.21	59.21	59.21
Omitted Group Mean (Pre-Incentives)		1.9	2.19		61.83	67.95
Total Effect (T × During + Inter.)		0.20	0.14, 0.36		4.87	2.89, 8.66
p-val Total Effect = 0		0.11	0.29, 0.02		0.20	0.48, 0.08
Treatment Effect %	9	11	7, 19	6	8	5, 15
Worker, Week FE	X	X	X	X	X	X
Day-of-week FE	X	X	X	X	X	X
N Worker-Days	17705	17705	15889	17705	17705	15889

Notes: Data at worker-day level. Outcomes coded 0 if worker was absent. Standard errors at clustered at the firm level. Total Effect reports the lincom of T × During plus the listed interaction (Late, Very Late shown comma-separated). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A4: Treatment Effects on Coordination - Firm Level

	Mins between 1st and last worker	**Mins between the 10th and 90th pctile worker**	**Deviation from start-time: 90th pctile worker**
	(1)	(2)	(3)
T × During	-22.07**	-9.53*	-13.89***
	[10.79]	[5.28]	[4.93]
Control Mean (Pre-Incentives)	129.38	84.4	47.45
Control Mean (Incentive Weeks)	115.39	70.62	31.38
Treatment Effect %	-17	-11	-29
Firm, Week FE	X	X	X
Day-of-week FE	X	X	X
N Firm-Days	2208	2970	2970

Notes: This table reports firm-day level outcomes on coordination in worker arrival times. All specifications include firm, day-of-week, phase by week, phase by strata fixed effects. Standard errors are clustered at the firm level. All outcome variables are in minutes, and are only computed for firm-days where strictly greater than 1 worker was present.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A5: Weekly Worker Earnings

	Weekly Earnings (INR)					
	excl. Incentives			incl. Incentives		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment × During	353.88	151.99	346.54	861.11***	680.28*	942.01**
	[317.81]	[362.87]	[395.35]	[320.99]	[368.65]	[399.68]
Treatment × During × Female		360.23			322.65	
		[233.40]			[239.35]	
Treatment × During × Late			-104.07			-164.41
			[364.01]			[374.22]
Treatment × During × Very Late			133.86			-97.14
			[315.10]			[323.39]
Control Mean (Pre-Incentives)	3496.51	3496.51	3496.51	3496.51	3496.51	3496.51
Omitted Group Mean (Pre-Incentives)		4041.7	4045.79		4041.7	4045.79
Total Effect (T × During + Inter.)		512.22	242.47, 480.40		1002.93	777.59, 844.86
p-val Total Effect = 0		0.11	0.52, 0.13		0.00	0.04, 0.01
Treatment Effect %	10	15	7, 14	25	29	22, 24
Worker, Week FE	X	X	X	X	X	X
N Worker-Weeks	2667	2667	2602	2667	2667	2602
N Firms	93	93	92	93	93	92

Notes: SEs clustered at firm level. All specifications include worker, phase by week, phase by strata fixed effects. This regression excludes owners/co-owners who also do stitching work at the firm. This regression excludes days of firm closure if no production took place at the firm. Earnings are coded as 0 for workers who are absent on a given day, and are winsorized at the 95th percentile. Total Effect reports the lincom of T × During plus the listed interaction (Late, Very Late shown comma-separated).
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A6: Pr. doing non production tasks

	$\mathbb{I}$\{Did any non-production tasks at the firm\}		
	(1)	(2)	(3)
Treatment × During	-0.03	0.01	-0.01
	[0.03]	[0.03]	[0.04]
Treatment × During × Female		-0.06**	
		[0.03]	
Treatment × During × Late			-0.04
			[0.03]
Treatment × During × Very Late			-0.01
			[0.04]
Control Mean (Pre-Incentives)	.16	.16	.16
Omitted Group Mean (Pre-Incentives)		.16	.16
Total Effect (T × During + Inter.)		-0.05	-0.05, -0.02
p-val Total Effect = 0		0.14	0.15, 0.64
Treatment Effect %	-16	-33	-33, -12
Worker, Week FE	X	X	X
Day-of-week FE	X	X	X
N Worker-Days	12465	12465	12333

Notes: SEs clustered at firm level. All specifications include worker, day-of-week, phase by week, phase by strata fixed effects. This regression excludes owners/co-owners who also do stitching work at the firm. This regression excludes days of firm closure if no production took place at the firm. Total Effect reports the lincom of T × During plus the listed interaction (Late, Very Late shown comma-separated).
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A7: Treatment Effects on Helper Attendance - Firm Level

	Fraction Helpers Present
	(1)
Treatment × During Incentives	0.03
	[0.04]
Control Mean (Pre-Incentives)	.83
Control Mean (Incentive Weeks)	.81
Treatment Effect %	4
Firm, Week FE	X
Day-of-week FE	X
N Firm-Days	3060

Notes: This table reports firm-day level outcomes on helper attendance. All specifications include firm, day-of-week, phase by week, phase by strata fixed effects. Standard errors are clustered at the firm level.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A8: Self-Reported Non-Labor Costs at Endline

	Electricity	**Thread, Needles**	**Transport**	**Repair, Maintenance**
	(1)	(2)	(3)	(4)
Treatment	-292.17	-1990.74	-1476.77*	71.79
	[599.90]	[1404.10]	[825.85]	[583.72]
Control Mean	5156.58	10438.93	4768.89	2240.91
Treatment Effect %	-5.67	-19.07	-30.97	3.2
N Firms	91	90	91	89

Notes: All specifications control for baseline costs and strata fixed effects. Robust standard errors reported.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A9: Purchased new inputs versus used existing inputs

	Purchased new inputs	**Used existing inputs**
	(1)	(2)
Treatment	-0.19	0.24
	[0.16]	[0.16]
Control Mean	.76	.24
Observations	34	34

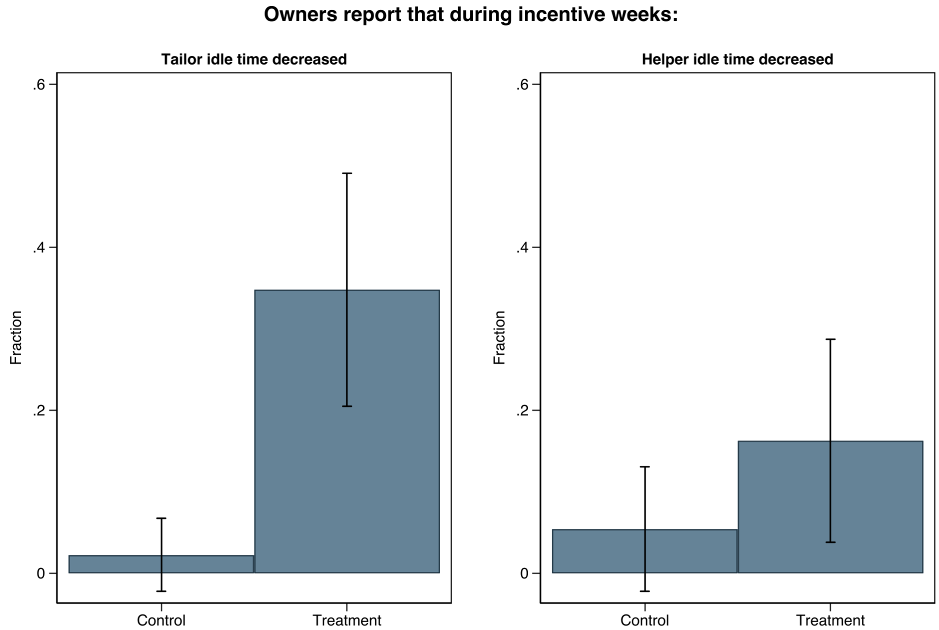
Notes: Round 3 firms only. "Question: In the last 4 weeks, did you purchase new inputs (thread/cones/needles), or did you use existing inputs (thread/cones/needles)?"

Table A10: Flexibility by Firm Size

	Small Firms		Large Firms	
	Mean	SD	Mean	SD
Pay Structure				
Piece Rate	0.90	(0.30)	0.60	(0.49)
Shift Rate	0.06	(0.23)	0.40	(0.49)
Average Monthly Earnings (Rs.)	16087.43	(5899.93)	16713.75	(5323.63)
Work Timings				
Timings Same	0.49	(0.50)	0.77	(0.42)
Timings Strict	0.15	(0.36)	0.50	(0.51)
Entry time is recorded	0.03	(0.17)	0.50	(0.51)
Pay cut if late	0.06	(0.23)	0.40	(0.49)
Margin of Lateness				
Acceptable to be 5 minutes late	0.97	(0.17)	0.94	(0.24)
Acceptable to be <= 30 minutes late	0.91	(0.29)	0.69	(0.47)
Acceptable to be > 30 minutes late	0.85	(0.36)	0.62	(0.49)
Observations	70		48	

Notes: SD in parenthesis (even columns). Small firms are defined as firms with up to 20 workers, and where the owner is often both a tailor (either full or part-time) and manager of the firm. Large firms include firms with at least 50 workers, where ownership and factory-floor management are separate.

Figure A1: Self-Reported Idle Time among Workers



95% CIs reported